Inverse Kinematics of Robotic Arm Manipulators

Zachary Cheng

May 18, 2025

1 Introduction

Inverse kinematics (IK) is the process of calculating the joint parameters (such as angles and positions) needed to place the end-effector (e.g. hand, gripper, foot) of a robot or animated character at a desired position and orientation in space. It plays a critical role in both robotics and computer animation. In these applications, it is generally easy for users to define the desired positions or trajectories, but difficult to directly give the values of the control variables.

2 General models

Consider a general form of robot whose control variables are denoted by a vector $\vec{c}$. An example is the 2-joint robotic arm illustrated in Fig. 1. Its state can be described by the angles of two arms, q_1 and q_2:

$$\vec{s} = \begin{pmatrix} q_1 \\ q_2 \end{pmatrix} \tag{1}$$

It has 2 control variables, which control the change of its state. The two variables are thus the angular velocities of the two joints,

$$\omega_i = \frac{dq_i}{dt}, i = 1, 2 \tag{2}$$

And the control vector is given by

$$\vec{c} = \begin{pmatrix} \omega_1 \\ \omega_2 \end{pmatrix} \tag{3}$$

Note that the control vector may be time varying. In general, the change of state of a robot underlying a control vector $\vec{c}(t)$ is given by

$$\frac{d\vec{s}(t)}{dt} = \vec{F}\left(\vec{s}(t), \vec{c}(t)\right) \tag{4}$$

The treatment of this general problem requires the calculus of variations, which is beyond the scope of this project. So we will limit our discussion to the situations where $\vec{c}$ is a constant, and the effect of $\vec{s}$ and $\vec{c}$ in$\vec{F}$can be separated. That is, we will only discuss the situations in which the state of a system can be written as

$$\vec{s}(t) = F\left(\vec{c}, t\right) \tag{5}$$

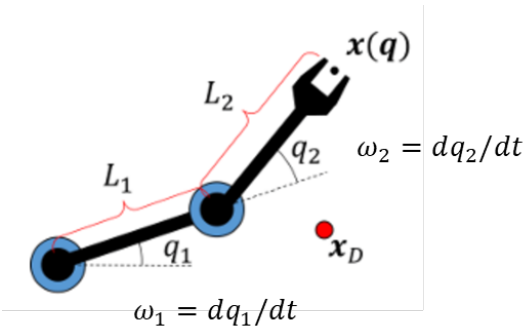

Figure 1: 2-joint robotic arm with 2 control variables

2.1 The Jacobian treatment in IK

The Jacobian of an arbitrary vector function $\vec{y} = f(\vec{x})$, with $f : R^n \to R^m$, is given as the following matrix of partial derivatives:

$$J(\vec{x}) = \frac{\partial f}{\partial \vec{x}} = \begin{pmatrix} \frac{\partial y_1}{\partial x_1} & .. & \frac{\partial y_1}{\partial x_n} \\ . & . & . \\ \frac{\partial y_m}{\partial x_1} & .. & \frac{\partial y_m}{\partial x_n} \end{pmatrix} \tag{6}$$

The (i, j) entry of the Jacobian indicates the rate of change of y_i with respect to the change of x_j. In other words, at a given vector $\vec{x}$, if there is a small change of $d\vec{x}$, the corresponding change of $\vec{y}$ is given by

$$d\vec{y} = J(\vec{x})\, d\vec{x} \tag{7}$$

$$\begin{pmatrix} dy_1 \\ .. \\ dy_m \end{pmatrix} = \begin{pmatrix} \frac{\partial y_1}{\partial x_1} & .. & \frac{\partial y_1}{\partial x_n} \\ . & . & . \\ \frac{\partial y_m}{\partial x_1} & .. & \frac{\partial y_m}{\partial x_n} \end{pmatrix} \begin{pmatrix} dx_1 \\ .. \\ dx_n \end{pmatrix} \tag{8}$$

Jacobian is the foundation of optimization techniques. Apply it to our IK problem, Eq. (5). Suppose we want to find $\vec{c}$ such that at time t, the system reaches the target state $\vec{s}(t) = \vec{s_t}$.

Now we know the Jacobian of the state vector with respect to the control vector:

$$J(\vec{c}) = \frac{\partial F}{\partial \vec{c}} = \begin{pmatrix} \frac{\partial s_1}{\partial c_1} & .. & \frac{\partial s_1}{\partial c_n} \\ . & . & . \\ \frac{\partial s_m}{\partial c_1} & .. & \frac{\partial s_m}{\partial c_n} \end{pmatrix} \tag{9}$$

We can pick an initial $\vec{c_0}$, calculate the error between the resulted $\vec{s_0}$ and the target $\vec{s_t}$:

$$\begin{pmatrix} \Delta s_1 \\ .. \\ \Delta s_m \end{pmatrix} = \vec{s_t} - F(\vec{c_0}) \tag{10}$$

From the physical significance of Jacobian, given by Eq. (8), we can compute the change of $\vec{c}$ to eliminate the error:

$$\begin{pmatrix} \Delta c_1 \\ .. \\ \Delta c_n \end{pmatrix} = \begin{pmatrix} \frac{\partial s_1}{\partial c_1} & .. & \frac{\partial s_1}{\partial c_n} \\ . & . & . \\ \frac{\partial s_m}{\partial c_1} & .. & \frac{\partial s_m}{\partial c_n} \end{pmatrix}^{-1} \begin{pmatrix} \Delta s_1 \\ .. \\ \Delta s_m \end{pmatrix} \tag{11}$$

When $m \neq n$, the operator $^{-1}$ is an pseudoinverse. Of course, the application of Jacobian is only valid when the change is small. So we usually use a learning rate to control the change of $\vec{c}$:

$$\begin{pmatrix} \Delta c_1 \\ .. \\ \Delta c_n \end{pmatrix} = \epsilon \begin{pmatrix} \frac{\partial s_1}{\partial c_1} & .. & \frac{\partial s_1}{\partial c_n} \\ . & . & . \\ \frac{\partial s_m}{\partial c_1} & .. & \frac{\partial s_m}{\partial c_n} \end{pmatrix}^{-1} \begin{pmatrix} \Delta s_1 \\ .. \\ \Delta s_m \end{pmatrix} \tag{12}$$

where $0 < \epsilon < 1$ is the learning rate. This technique is known as Gradient Descent, because Jacobian actually gives the gradients of s_i with respect to $\vec{c}$. To make the calculation stable, various techniques have been developed, such as stochastic gradient descent, batch normalization, etc.

2.2 Solution to the problem presented in the simulation

The simulation (`https://www.geogebra.org/m/hunha83a`) presents a problem concerning a two-joint robotic arm that is attempting to "throw" a point to hit a target.

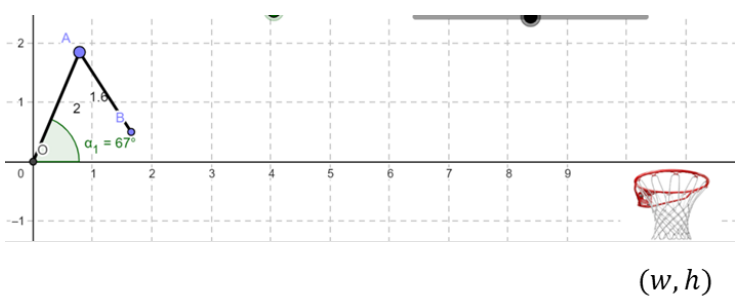

Figure 2: 2-joint robotic arm throwing a ball to a target hoop

In this case, when ω_1 and ω_2 are constant, the forward kinematics (FK) problem is formulated below. Assume that their initial angles are α_1 and α_2, respectively. At time t,

$$q_1 = \alpha_1 + \omega_1 t \tag{13}$$

$$q_2 = \alpha_2 + \omega_2 t \tag{14}$$

Without loss of generality, the position of the 1st joint is fixed at $(0, 0)$.The position of the 2nd joint is

$$\vec{r_2} = \begin{pmatrix} L_1 \cos(\alpha_1 + \omega_1 t) \\ L_1 \sin(\alpha_1 + \omega_1 t) \end{pmatrix} \tag{15}$$

The position of the gripper is given by

$$\vec{r}(t) = \vec{r_2} + \begin{pmatrix} L_2 \cos(q_1 + q_2) \\ L_2 \sin(q_1 + q_2) \end{pmatrix} \tag{16}$$

$$= \begin{pmatrix} L_1 \cos(\alpha_1 + \omega_1 t) \\ L_1 \sin(\alpha_1 + \omega_1 t) \end{pmatrix} + \begin{pmatrix} L_2 \cos(\alpha_1 + \alpha_2 + (\omega_1 + \omega_2) t) \\ L_2 \sin(\alpha_1 + \alpha_2 + (\omega_1 + \omega_2) t) \end{pmatrix} \tag{17}$$

If the gripper releases a ball at this time, the velocity of the ball is the same as the velocity of the gripper, which is given by

$$\vec{v} = \vec{v_1} + \vec{v_{21}} \tag{18}$$

or we can simply take the derivative of $\vec{r}$:

$$\vec{v}(t) = \frac{d\vec{r}}{dt} = \omega_1 \begin{pmatrix} -L_1 \sin(\alpha_1 + \omega_1 t) \\ L_1 \cos(\alpha_1 + \omega_1 t) \end{pmatrix} + (\omega_1 + \omega_2) \begin{pmatrix} -L_2 \sin(\alpha_1 + \alpha_2 + (\omega_1 + \omega_2) t) \\ L_2 \cos(\alpha_1 + \alpha_2 + (\omega_1 + \omega_2) t) \end{pmatrix} \tag{19}$$

After releasing, the ball follows this kinematics equation:

$$\vec{r}(t) = \vec{r_0} + \vec{v_0} t + \frac{1}{2} \vec{g} t^2 \tag{20}$$

where $\vec{g} = (0, -g)$ is the gravitational acceleration, and $\vec{r_0}$, $\vec{v_0}$ are the position and velocity of the ball when being released.

In order for the ball to hit a target at the position $\vec{p} = (w, h)$, there must exist a time τ such that

$$\vec{r_0} + \vec{v_0} \tau + \frac{1}{2} \vec{g} \tau^2 = \vec{p} \tag{21}$$

Denote

$$\vec{r_0} = \begin{pmatrix} r_{0x} \\ r_{0y} \end{pmatrix}, \vec{v_0} = \begin{pmatrix} v_{0x} \\ v_{0y} \end{pmatrix}$$

Then

$$\tau = \frac{w - r_{0x}}{v_{0x}} \tag{22}$$

$$r_{0y} + v_{0y} \tau - \frac{1}{2} g \tau^2 = h \tag{23}$$

or

$$r_{0y} + v_{0y} \frac{w - r_{0x}}{v_{0x}} - \frac{1}{2} g \left(\frac{w - r_{0x}}{v_{0x}} \right)^2 = h \tag{24}$$

Combining Eq (17), (19) and (24), the IK problem is to find the angular velocity $\omega_1$and ω_2, such that at a user-specified time t_0, the following equations and constraints are satisfied:

$$\vec{r_0} = \begin{pmatrix} r_{0x} \\ r_{0y} \end{pmatrix} = \begin{pmatrix} L_1 \cos(\alpha_1 + \omega_1 t_0) \\ L_1 \sin(\alpha_1 + \omega_1 t_0) \end{pmatrix} + \begin{pmatrix} L_2 \cos(\alpha_1 + \alpha_2 + (\omega_1 + \omega_2) t_0) \\ L_2 \sin(\alpha_1 + \alpha_2 + (\omega_1 + \omega_2) t_0) \end{pmatrix} \tag{25}$$

$$\vec{v_0} = \begin{pmatrix} v_{0x} \\ v_{0y} \end{pmatrix} = \omega_1 \begin{pmatrix} -L_1 \sin(\alpha_1 + \omega_1 t_0) \\ L_1 \cos(\alpha_1 + \omega_1 t_0) \end{pmatrix} + (\omega_1 + \omega_2) \begin{pmatrix} -L_2 \sin(\alpha_1 + \alpha_2 + (\omega_1 + \omega_2) t_0) \\ L_2 \cos(\alpha_1 + \alpha_2 + (\omega_1 + \omega_2) t_0) \end{pmatrix} \tag{26}$$

$$r_{0y} + v_{0y}\frac{w - r_{0x}}{v_{0x}} - \frac{1}{2}g\left(\frac{w - r_{0x}}{v_{0x}}\right)^2 = h \tag{27}$$

Note that there is only 1 constraint, namely Eq. (27), whereas there are 2 unknown variables, $\omega_1$and ω_2. Therefore, there are an infinite number of solutions. Other constraints can be applied to remove physically inappropriate solutions. For example, we can require that the ball is falling when it hits the target. That is,

$$v_{0y} - g\tau < 0 \implies v_{0y} - g\frac{w - r_{0x}}{v_{0x}} < 0 \tag{28}$$

Also note that the equations are nonlinear, so analytical solutions are usually not available. Numerical methods, such as Gradient Descent (GD), are employed to get solutions. Many of such methods depend on the Jacobian treatment outlined in Section 2.1, particularly Eq. (12).